

APPENDIX

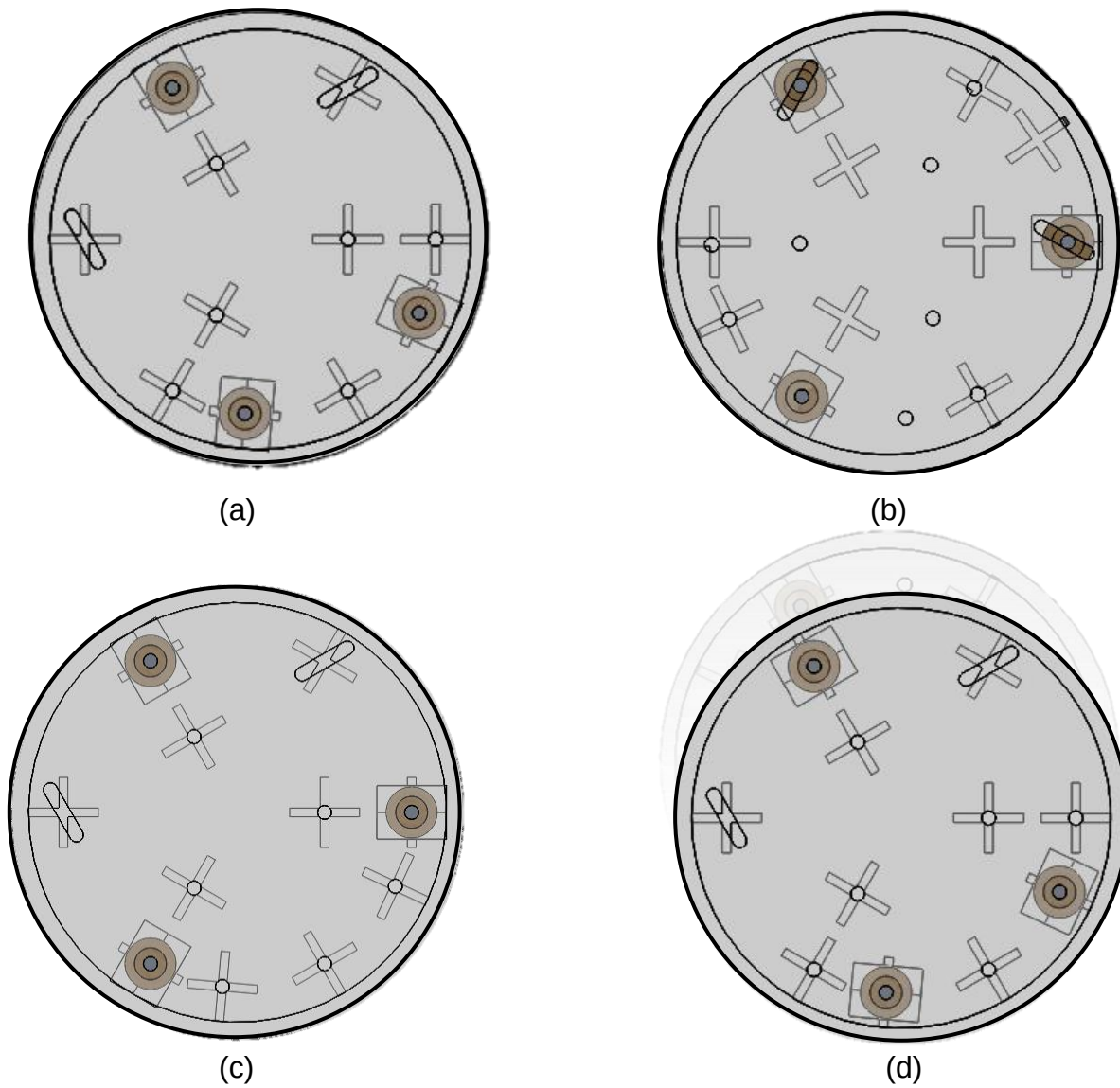


Figure 4 Examples of some of the configurations available in the kinematic coupling
(a) Stable Isosceles (b) "Adjustable Error" Sliding Position
(c) Stable Equilateral (d) Unstable Equilateral

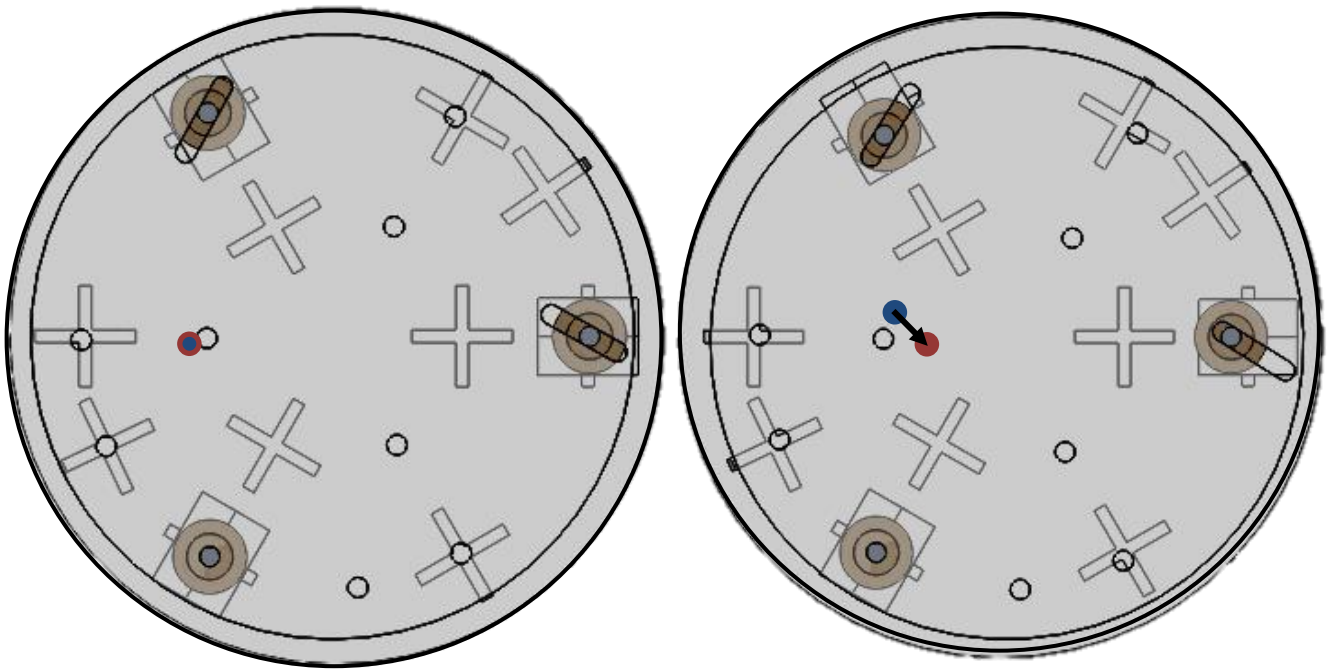


Figure 5. Demonstration of positional inaccuracy of learning tool

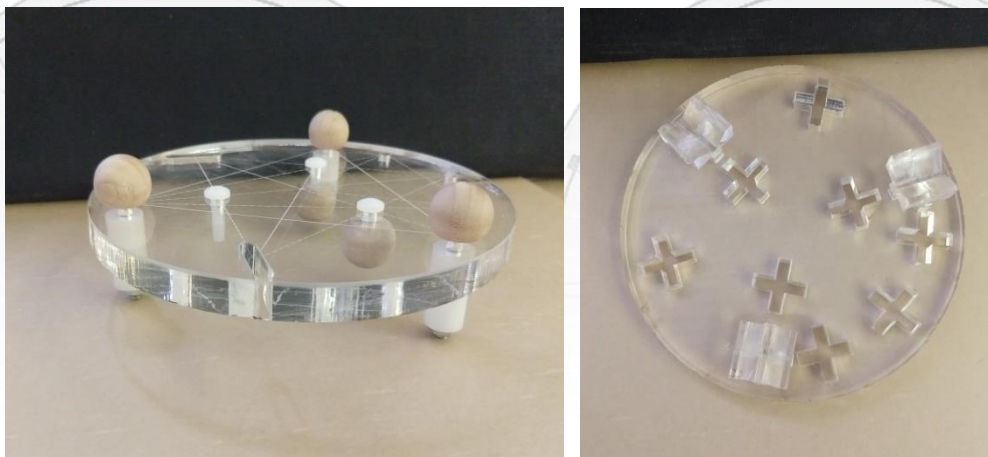


Figure 7. Top and Bottom plate assemblies of the kinematic coupling

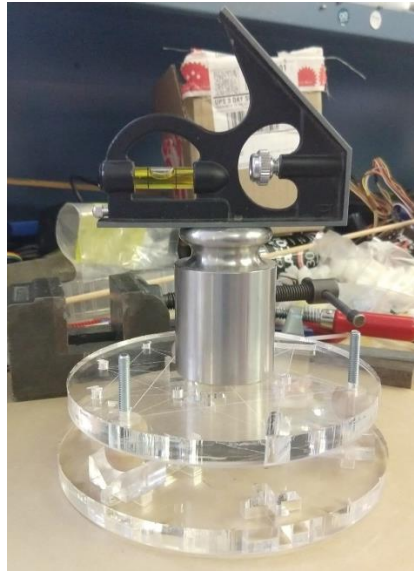
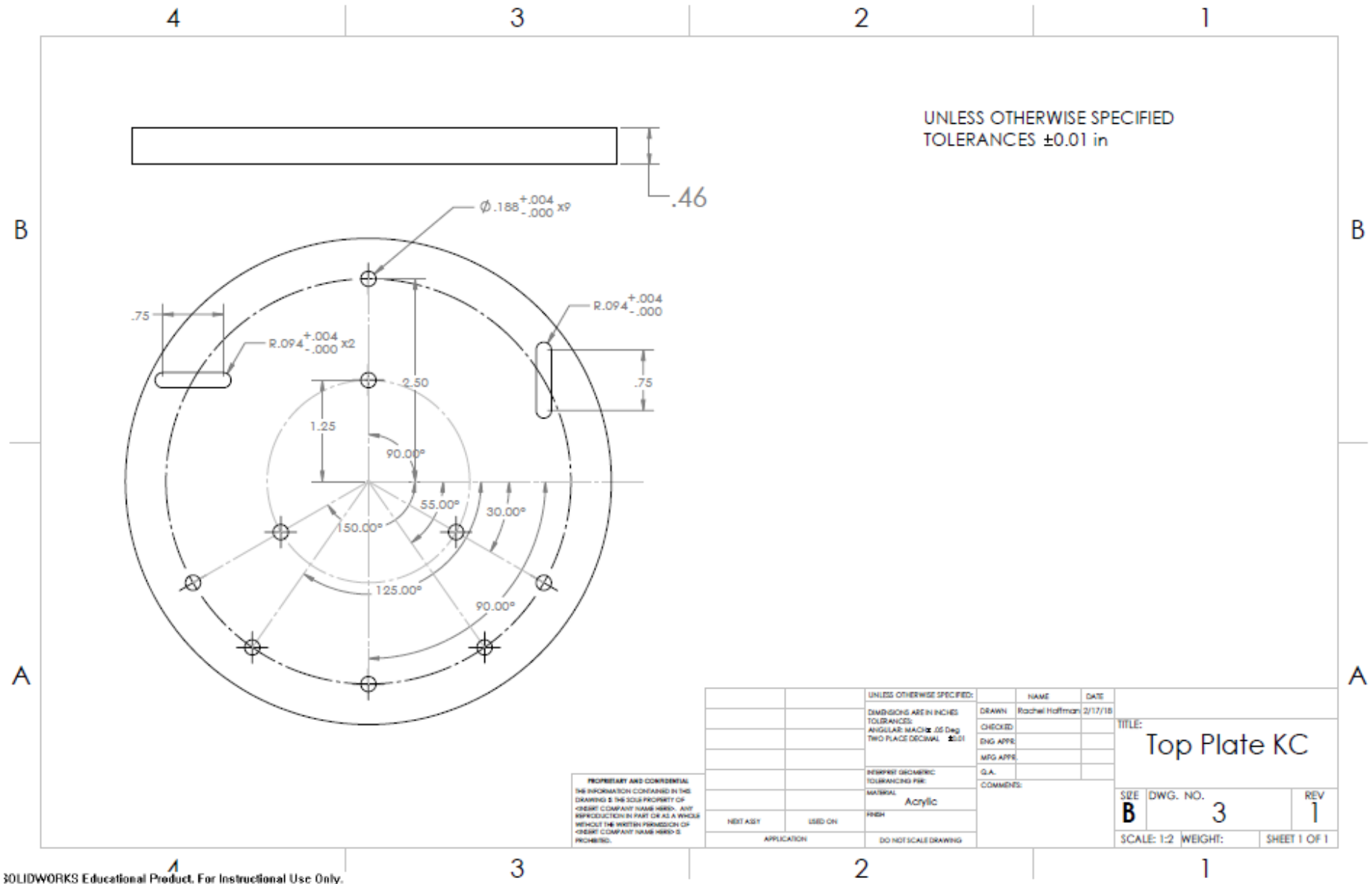
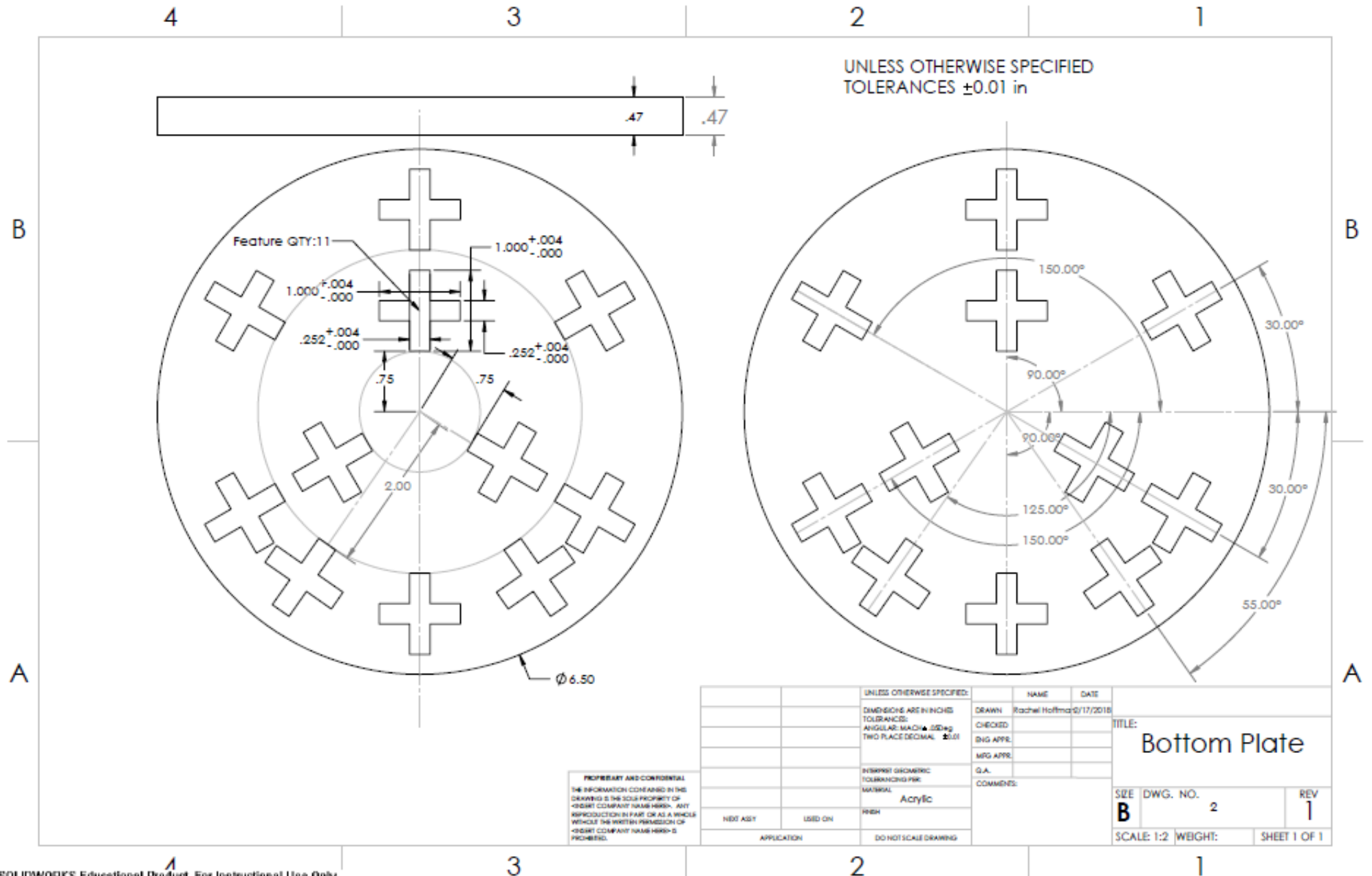
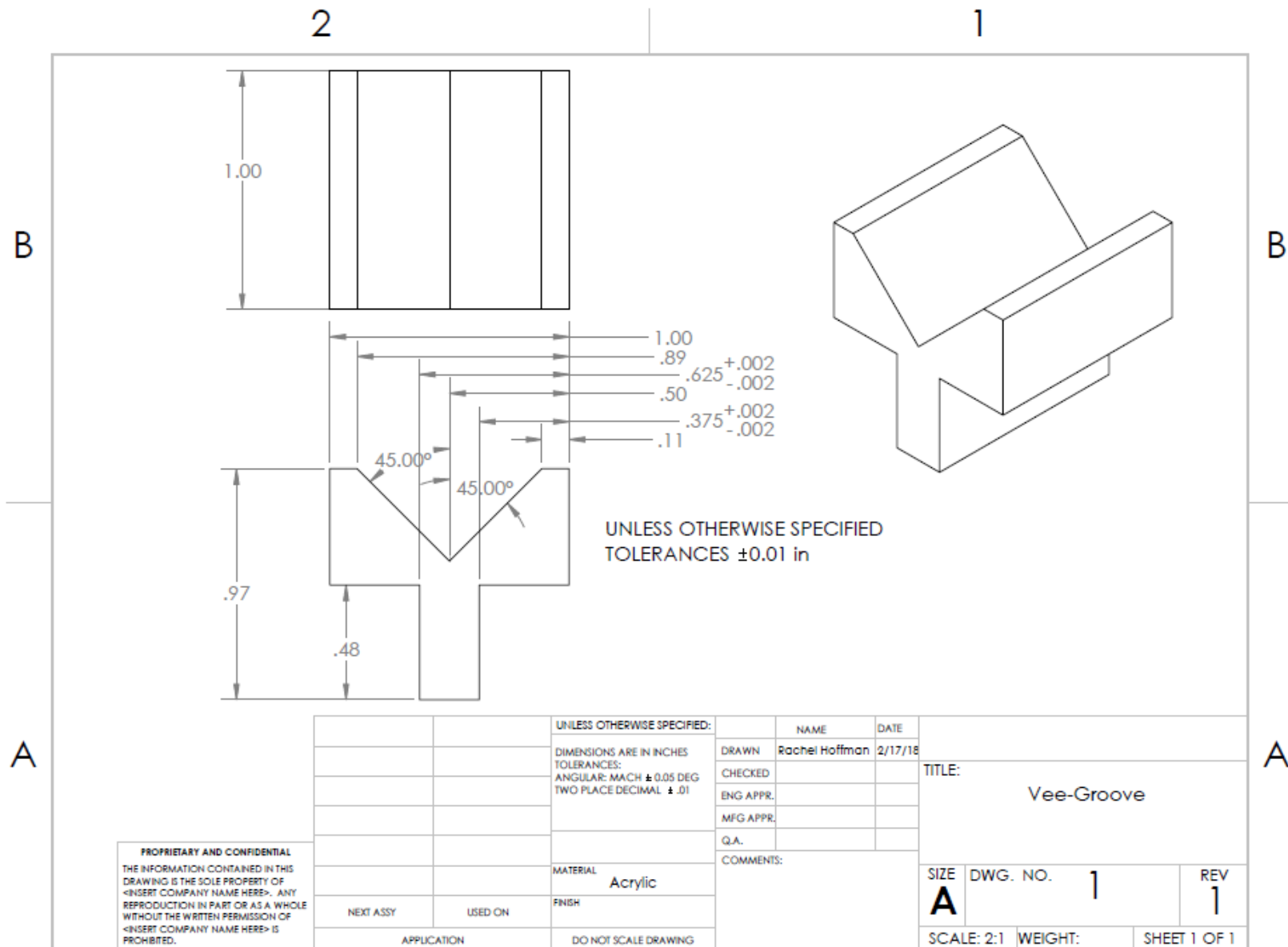


Figure 8. Testing the tilt of the kinematic coupling with a level







```
%Kinematic Coupling Measured Data
```

```
%1inch = 25400um
```

```
clear all
```

```
close all
```

```
HalfKGdata=[ 0 .00025 .00025 .00025 .00045 .0002 .00025 .00025 .00025 .0001];
```

```
OneKG_ZeroFromHalf=[-.0011 -.00125 -.0015 -.0015 -.0011 -.0014 -.0014 -.0012 -.0014];
```

```
OneKG=[.0005 .0005 .0005 .0005 .0004 .0004];
```

```
AvgHalfKg=mean(HalfKGdata*25400);
```

```
StdHalfKg=std(HalfKGdata*25400);
```

```
AvgOneKg_ZeroFromHalf=mean(OneKG_ZeroFromHalf*25400);
```

```
StdOneKg_ZeroFromHalf=std(OneKG_ZeroFromHalf*25400);
```

```
AvgOneKg=mean(OneKG*25400);
```

```
StdOneKg=std(OneKG*25400);
```

```
figure
```

```
plot(HalfKGdata*25400,'ro')
```

```
hold on
```

```
plot(OneKG_ZeroFromHalf*25400,'k*')
```

```
xlabel('Trial')
```

```
ylabel('Displacement, um')
```

```
legend('0.5 kg load', '1kg Load')
```

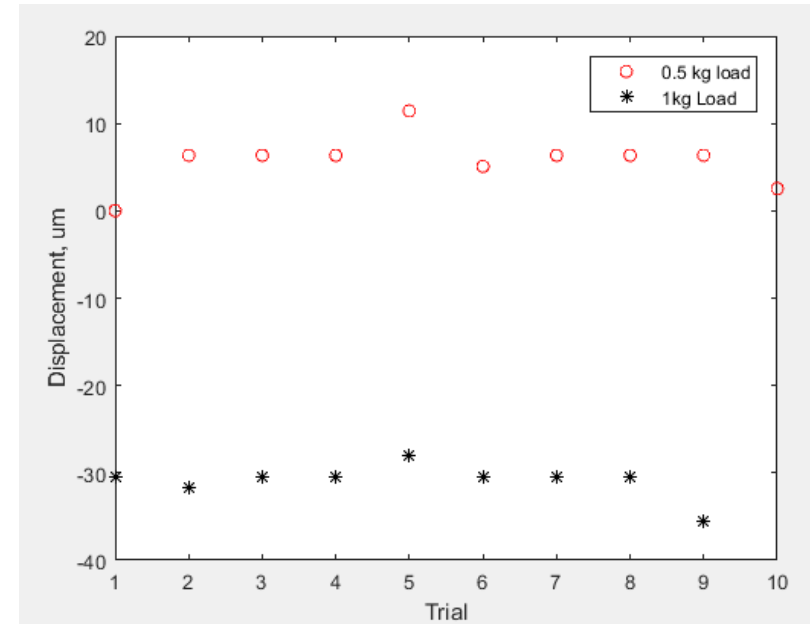
```
[AvgHalfKg 2*StdHalfKg;AvgOneKg_ZeroFromHalf 2*StdOneKg_ZeroFromHalf;AvgOneKg 2*StdOneKg ]
```

```
%F=kx
```

```
k=(0.5*9.8)/(abs(AvgOneKg_ZeroFromHalf-AvgHalfKg))
```

```
yaxis=[.0002 .0001 0 0 .0002 0 .00025 .00025 0 .0001];
```

```
stdy=2*std(yaxis*25400)
```



Plate_Circular_Central.xls

To determine deflection of a centrally loaded circular plate
By Alex Slocum, 1/1/04, last modified 09/21/04 by Xue'en Yang
Enters numbers in **BOLD**, Results in **RED**

Schematic

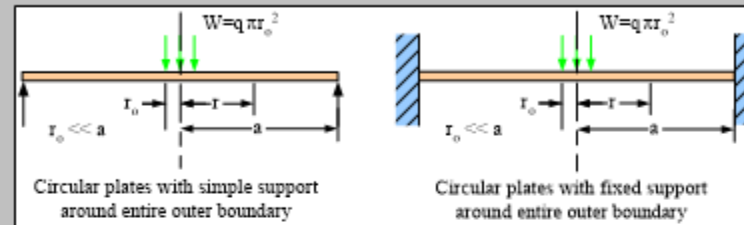


Plate dimensions and properties	Values	ProE Simulation
Radius, a (mm)	150	150
Thickness, t (mm)	12.70	1.50
Modulus of elasticity, E (N/mm ²)	3000	200000
Poisson ratio, v	0.37	0.29
Maximum stress, sigmax (N/mm ²)	70	1000
Total area	70686	
Loading		
loading pressure, q (N/mm ²)	0.0049	1
Outer radius of loading, ro (mm) (must be <<a)	25.4	10
Outer radius used in calculation, ro_prime (mm)	25.40	
Total load applied to the plate, W (N)	10	
Outer boundary support:		
Simply supported:		
Center displacement, dc (mm)	-0.018	-18.43084843
Slope at outer boundary, ta (radians)	0.000146	0.008365 deg
Bending moment at center, Mc (N-mm/mm)	8	
Bending stress at center, sigss (N/mm ²)	0	
Safety factor (max allowable stress/bending stress)	221.86	
Fixed		
Center displacement, fdc (mm)	-0.007	-2.238
Max slope (@ r=0.368a) (radians)	0.000	
Radius at maximum slope (mm)	55.2	
Bending moment at center, fMc (N-mm/mm)	2	
Bending moment at outer fixed edge fMa (N-mm)	-1	
Bending stress at center, sigfi (N/mm ²)	0	
Bending stress at outer fixed edge, sigfo (N/mm ²)	0	
Safety factor (max allowable stress/bending stress)	978.62	

Kinematic Coupling Calculations_RM1.xls									
2/13/2018									
http://pergatory.mit.edu/kinematiccouplings/doc									
Enter Use Values Calculated Values									
To Design and Experiment with Different Parameters of Ball-Vee Maxwell Couplings									
System Geometry Assuming Spheres and Vees with equal preload									
1	Dball (mm)	12.7	Ball Diameter						
2	Rball (mm)	6.35	Ball Radius						
3	Rgroove (mm)	1000000	Groove Diameter (Large for flat V)						
4	Dcoupling (mm)	150	Coupling Circle Diameter						
5	Rcoupling (mm)	75							
6			Preload Force over each ball (negative)						
7	XerrRep (mm)	0	X location of error reporting						
8	YerrRep (mm)	0	Y location of error reporting						
9	ZerrRep (mm)	0	Z location of error reporting						
10	BallMatYieldStress (Pa)	13800000							
11	BallMatYoungMod (Pa)	11000000000							
12	BallMatPoisson	0.3							
13	GrooveMatYieldStress (Pa)	138000000							
14	GrooveMatYoungMod (Pa)	11000000000							
15	GrooveMatPoisson	0.3							
16	AngleBetweenBalls	2.094395102	Standard is 120deg						
17	Groove Angle (rad)	0.785398163	Standard 90deg Vee = PI/4 (half of total angle)						
18	Applied Preload (Assuming Centrally Loaded)								
19	FPx (N)	0	Radial Position of Preload (RPL) (mm)	75					
20	FPy (N)	0	Angular Position of Preload (ThetaP)	1.5707963					
21	FPz (N)	-1.666666667							
22	Applied Load and its Location								
23	FLx (N)	0	XL (mm)	0					
24	FLy (N)	0	YL (mm)	0					
25	FLz (N)	-5	ZL (mm)	0					
26	CALCULATIONS								
27	Rball1	6.35E+00							
28	Rball2	6.35E+00							
29	Rball3	6.35E+00							
30	BallAng1	1.57E+00							
31	BallAng2	3.67E+00							
32	BallAng3	5.76E+00							
33	GrooveAng1	7.65E-01							
34	GrooveAng2	7.85E-01							
35	GrooveAng3	7.85E-01							
36	Dball	1.27E+01							
37	Coupling Locations and Directional Cosines								
38	First Ball (Contact points 1 and 2)			Second Ball (Contact points 3 and 4)			Third Ball (Contact points 5 and 6)		
39	XL1(m)	-4.43E-03	X2_3(m)	-6.72E-03	X3_5(m)	6.72E-03			
40	YL2(m)	-4.43E-03	X2_4(m)	-6.72E-03	X3_6(m)	6.72E-03			
41	YL1(m)	7.50E-03	Y2_3(m)	-3.38E-03	Y3_5(m)	-4.14E-03			
42	YL2(m)	7.50E-03	Y2_4(m)	-4.14E-03	Y3_6(m)	-3.38E-03			
43	ZL1(m)	-4.43E-03	Z2_3(m)	-4.43E-03	Z3_5(m)	-4.43E-03			
44	ZL2(m)	-4.43E-03	Z2_4(m)	-4.43E-03	Z3_6(m)	-4.43E-03			

35	DLttl		UJ			
36	Dchange in dist/ distance between balls					
37	DL_DD1		0			
38	DL_DD2		0			
39	DL_DD3		0			
40	Coupling Error Rotations					
41	EpsZ1		0			
42	EpsZ2		0			
43	EpsZ3		0			
44	EpsX		1.60915E-05			
45	EpsY		-9.29041E-06			
46	EpsZ		0			
47	HTM					
48			1	0	-9.29E-06	0
49			0	1	-1.61E-05	0
50		9.29041E-06		1.60915E-05	1	-2.401E-06
51			0	0	0	1
52						
53	POI		0	0	0	0
54	XYZInt		0			
55			0			
56			1			
57	Error motions at point of interest					
58	DeltaX		0			
59	DeltaY		0			
60	DeltaZ		-2.40078E-06			
61			1			
62	DeltaRMS		2.40078E-06			
63	DeltaRMSmm		0.002400779			
64						
65	RMS Applied Force		5			
66	RMS Stiffness		2.08265717			
67						
68						
69						
70						
71						
72						

Kinematic Coupling 3Groove Design.xls									
To design three groove kinematic couplings									
Written by Alex Slocum 4/1/1992. See Edit History worksheet									
Metric units only! Enters numbers in BOLD, Results in RED									
Standard 120 degree equal size groove coupling? (contact forces are inclined at 45 to the XY plane. For non standard designs, enter geometry after results section)						Material properties			
						material	aluminum		
						Yield stress		Mat Label	
System geometry (XY plane is assumed to contain the ball centers)						Acrylic	7.00E+07	1	
Dbeg (mm) =	18.796	Equivalent diameter ball to contact groove at same points				Wood	3.17E+07	2	
Rbminor (mm) =	9.40	"Ball" minor radius, = Dbeg/2 is spherical or crowned cone				CARBIDE	2.76E+09	3	
Rbmajor (mm) =	9.4	"Ball" major radius, = Dbeg/2 if spherical contact				user defined: RC 54 SS	5.52E+08	4	
Rgroove (mm) =	1.00E+06	Groove radius (negative for a trough)				Elastic modulus			
Costheta =	TRUE	Is ball major radius along groove axis?				Acrylic	3.20E+09	1	
Dcoupling (mm) =	100	Coupling diameter				Wood	1.00E+09	2	
Fpreload (N) =	5.00	Preload force over each ball				CARBIDE	3.10E+11	3	
Xerr (mm) =	100.0	X location of error reporting				user defined: RC 54 SS	2.04E+11	4	
Yerr (mm) =	0.0	Y location of error reporting				Poisson ratio			
Zerr (mm) =	36.0	Z location of error reporting				Acrylic	0.37	1	
Auto select material values (enter other_4 to the right)						Wood	0.3	2	
Matlabball =	2	Enter 1 for plastic, 2 for steel, 3 for carbide, 4 for user defined, 5 where each ball and groove is defined individually				CARBIDE	0.30	3	
Matlabgroove =	1					user defined: RC 54 SS	0.29	4	
Min. yield strength (Pa, psi)		3.17E+07		4,594					
Largest contact ellipse major diameter (mm)		-0.681							
Smallest contact ellipse major diameter (mm)		-0.680		0.004933813			4933.813103		
Largest contact stress ratio		-0.37							
RMS applied force (N)		5							
RMS stiffness (N/micron)		2.61							
Applied force's Z,Y,Z values and coordinates						Coupling centroid location			
FLx (N) =	0.00	XL (mm) =		0	xc (mm)	0.000		XL (m) =	0.000
FLy (N) =	0.00	YL (mm) =		0	yc (mm)	0.000		YL (m) =	0.000
FLz (N) =	5.00	ZL (mm) =		36	zc (mm)	0.000		ZL (m) =	0.036
Results: Hertz stresses and deformations									
Error displacements at the point of interest (micron)									
DeltaX (mm)	0.00E+00		DeltaY (mm)		0.00E+00		DeltaZ (mm)	1.92E-03	1.92E-03
Groove normal forces (N)		Contact pressure (Pa)		Max shear stress/(ult. tensile/2)		Deflection (+into ball)		Contact ellipse	
Fbnone	-4.7	sigone		-1.94E+07	-0.368	delone	2.14E-06	-3.41E-04	-3.40E-04
Fbntwo	-4.7	sigtwo		-1.94E+07	-0.368	deltwo	2.14E-06	-3.41E-04	-3.40E-04
Ball-Groove 2								Contact ellipse	
Groove normal forces		Contact stress				Deflection			
Fbnthree	-4.7	sigthree		-1.94E+07	-0.368	delthree	2.14E-06	-3.41E-04	-3.40E-04
Fbnfour	-4.7	sigfour		-1.94E+07	-0.368	delfour	2.14E-06	-3.41E-04	-3.40E-04
Ball-Groove 3								Contact ellipse	
Groove normal forces		Contact stress				Deflection			
Fbnfive	-4.7	sigfive		-1.94E+07	-0.368	delfive	2.14E-06	-3.41E-04	-3.40E-04
Fbnsix	-4.7	sigsix		-1.94E+07	-0.368	delsix	2.14E-06	-3.41E-04	-3.40E-04